OsmoFlux — Equation Reference

1D Axial Series-Current RED Model · Activity-Corrected Nernst EMF · Dual-Mechanism Fouling · Rohan Agarwal

1. EMF & Open-Circuit Voltage

Nernst EMF — per axial segment j

$$\mathcal{E}_j = \alpha \cdot \frac{2RT}{F} \ln\left(\frac{\gamma_{H,j}\, C_{H,j}}{\gamma_{L,j}\, C_{L,j}}\right)$$

α = ALPHA_PERM = 0.975 (Fumasep FKS-PET-130)
γ_H, γ_L = activity coefficients — Robinson & Stokes (1959), log-linear interp.
$C_{H,j}, C_{L,j}$ = segment concentrations from C_H_IN_MOL_L, C_L_IN_MOL_L
$j = 1\ldots$ N_SEG (default 40 segments per cell pair)

Stack open-circuit voltage [V_OC_EQ_V]

$$V_{\mathrm{OC}} = \sum_{j=1}^{N \times N_{\mathrm{seg}}} \mathcal{E}_j \,-\, N \cdot V_{\mathrm{elec}}$$

N = N_CELL_PAIRS (swept 2–80 in geometry optimisation)
$V_{\mathrm{elec}} = 0.1\,\mathrm{V}$ per electrode pair — Veerman (2010)

2. Internal Resistance & Power

Segment resistance (Thévenin equivalent)

$$R_j = \frac{\mathrm{ASR}_{\mathrm{CEM}} + \mathrm{ASR}_{\mathrm{AEM}}}{\beta_{\mathrm{mem}} \cdot A_{\mathrm{mem}}} + \frac{d_{\mathrm{sp}}}{\beta_{\mathrm{sol}} \cdot \sigma_j \cdot A_{\mathrm{mem}}}$$

ASR_CEM_OHM_M2 $= 3.6\times10^{-4}\,\Omega\cdot\mathrm{m}^2$ (FKS-PET-130 midpoint)
ASR_AEM_OHM_M2 $= 2.35\times10^{-4}\,\Omega\cdot\mathrm{m}^2$ (FAS-PET-130 midpoint)
$\beta_{\mathrm{mem}} = 1.10$, $\beta_{\mathrm{sol}} = 1.56$ — spacer shadow factors (Vermaas 2014)
σ_j = NaCl conductivity, log-log interp.; d_{sp} = GAP_M $= 2\times10^{-4}\,\mathrm{m}$

Stack internal resistance [R_INT_EQ_OHM]

$$R_{\mathrm{int}} = \sum_{j=1}^{N \times N_{\mathrm{seg}}} R_j$$

Series-current topology: one current I through all $N \times N_{\mathrm{seg}}$ segments

Maximum power density [P_DENS_W_M2]

$$P_{\mathrm{dens}} = \frac{V_{\mathrm{OC}}^2}{4\, R_{\mathrm{int}}\, N\, A_{\mathrm{mem}}}$$

Analytical result at matched load $R_{\mathrm{load}} = R_{\mathrm{int}}$
P_TOTAL_W = P_DENS_W_M2 × N_CELL_PAIRS × A_MEM_M2

3. Fouling Mechanisms

CDF — divalent ion scaling (inlet-heavy)

$$\Delta R_{\mathrm{CDF},j} = \beta \cdot f_{\mathrm{div}} \cdot \left(\frac{C_{H,j}}{C_{\mathrm{ref}}}\right)^n \cdot \mathrm{ASR}_{\mathrm{CEM}}$$

β = FOULING_BETA $\in [0,1]$; f_{div} = DIVALENT_FRACTION
C_{ref} = FOULING_C_REF_MOL_L $= 0.5\,\mathrm{mol/L}$; n = FOULING_N_EXP $= 0.5$
Source: Długołęcki (2010); penalty peaks at $j = 1$ (inlet) where C_H is highest

Exponential biofilm — outlet-heavy $\phi(j)$

$$\Delta R_{\mathrm{bio},j} = \beta \cdot \mathrm{ASR}_{\mathrm{CEM}} \cdot \frac{\exp\left(\frac{j-1}{N_{\mathrm{seg}}-1}\right) - 1}{e-1}$$

$\phi(j=1) = 0$ (no biofilm at inlet); $\phi(j = N_{\mathrm{seg}}) = \beta \cdot \mathrm{ASR}_{\mathrm{CEM}}$ (max at outlet)
Normalisation by $(e-1)$: boundary conditions satisfied exactly
Biofilm builds where ionic gradient is weakest (toward outlet)

Total segment resistance with fouling

$$R_j^{\mathrm{foul}} = R_j + \Delta R_{\mathrm{CDF},j} + \Delta R_{\mathrm{bio},j}$$

Combined penalty 4–7 pp below independent sum (sub-additive)
Mechanism: both terms compete for the same Thévenin driving force

4. Cascade Architecture & Harm Reduction

Harm reduction index HR

$$\mathrm{HR} = 1 - \frac{C_{H,\mathrm{out}} - C_{\mathrm{ref,sea}}}{C_{H,\mathrm{in}} - C_{\mathrm{ref,sea}}}$$

$C_{H,\mathrm{out}}$ = brine salinity leaving RED system [mol/L]
$C_{H,\mathrm{in}}$ = C_H_IN_MOL_L (raw brine inlet)
$C_{\mathrm{ref,sea}}$ = ambient receiving-body salinity [mol/L]
HR = 1 → full compliance; HR = 0 → no improvement

Salt extracted per cascade stage s

$$\dot{m}_s = I_s \cdot N_{\mathrm{cp},s} \cdot \frac{M_{\mathrm{NaCl}}}{F}$$

$I_s = V_{\mathrm{OC},s}/(2R_{\mathrm{int},s})$ = optimal current [A]
$N_{\mathrm{cp},s}$ = N_CELL_PAIRS in stage s; $M_{\mathrm{NaCl}} = 58.44\,\mathrm{g/mol}$

Cascade power gain G vs single stack

$$G = \frac{P_{\mathrm{cascade}} - P_{\mathrm{single}}}{P_{\mathrm{single}}} \times 100\%$$

Positive only above C_H_IN_MOL_L $\gtrsim 1.5\,\mathrm{mol/L}$
Up to +418% at hypersaline ($C_H = 3.0$, $S = 10$, FOULING_BETA $= 0.8$)
Fouling shift: optimal N_CELL_PAIRS per stage falls as FOULING_BETA rises

5. Techno-Economic Analysis

Net present value (primary metric)

$$\mathrm{NPV} = \sum_{t=1}^{T} \frac{R_t - C_t}{(1+r)^t} - C_{\mathrm{CAPEX}}$$

R_t = electricity revenue + avoided brine disposal cost ($\sim\$0.04/\mathrm{m}^3$)
Fresh-dilute cascade: P(NPV > 0) $= 36.1\%$, median $= -\$37.0\,\mathrm{M}$ (10,000-sample Monte Carlo)

LCOE (reference only — not primary metric)

$$\mathrm{LCOE} = \frac{C_{\mathrm{CAPEX}} + \sum C_t (1+r)^{-t}}{\sum E_t (1+r)^{-t}}$$

P50 = \$3.05/kWh — misleading without disposal credit
NPV is the correct metric; value driven by avoided disposal cost

6. 1D Series-Current Solver

Axial salt conservation (per segment j)

$$C_{H,j+1} = C_{H,j} - \frac{\dot{n}_j}{Q_H} \qquad \dot{n}_j = \frac{I \cdot N_{\mathrm{cp}}}{F \cdot N_{\mathrm{seg}}}$$

I = single stack current shared by all $N_{\mathrm{cp}} \times N_{\mathrm{seg}}$ segments (series topology)
$\dot{n}_j$ = molar salt transfer rate at segment j [mol/s]
Q_H = volumetric brine flow rate (Q_H_M3_S) [m³/s]

Max-power condition (bisection root)

$$f(I) = E_{\mathrm{total}}(I) - 2\, I\, R_{\mathrm{total}}(I) = 0$$

$E_{\mathrm{total}}(I) = \sum_j \mathcal{E}_j(C_{H,j}(I), C_{L,j}(I))$ — depends on I via salt depletion
$R_{\mathrm{total}}(I) = \sum_j R_j^{\mathrm{foul}}(C_{H,j}(I))$ — depends on I via local concentrations
Solved by bisection over $I \in [0, I_{\max}]$; N_SEG = 40 axial segments

Optimal current and power at solution

$$I^* = \frac{E_{\mathrm{total}}(I^*)}{2\, R_{\mathrm{total}}(I^*)} \qquad P_{\max} = \frac{E_{\mathrm{total}}^2}{4\, R_{\mathrm{total}}}$$

Non-trivial: E_{total} and R_{total} both vary with I (salt depletion couples them)
Reduces to $P = V_{\mathrm{OC}}^2/(4R_{\mathrm{int}})$ only when concentrations are I-independent

Robinson & Stokes (1959) · Długołęcki et al. (2010) J. Membr. Sci. 346 · Vermaas et al. (2014) · Veerman et al. (2010) · Fumasep FKS-PET-130 / FAS-PET-130 datasheets